

DATASET : It is a seeds dataset from kaggle. It is a classification dataset with 3 species and exactly 4 numerical features, so it fits our 4-2-3 neural network.

INPUT LAYER : 4 Neurons

$$X = [15.26, 14.84, 0.871, 5.763]$$

$$X_1 = \text{Area}$$

$$X_2 = \text{perimeter}$$

$$X_3 = \text{compactness}$$

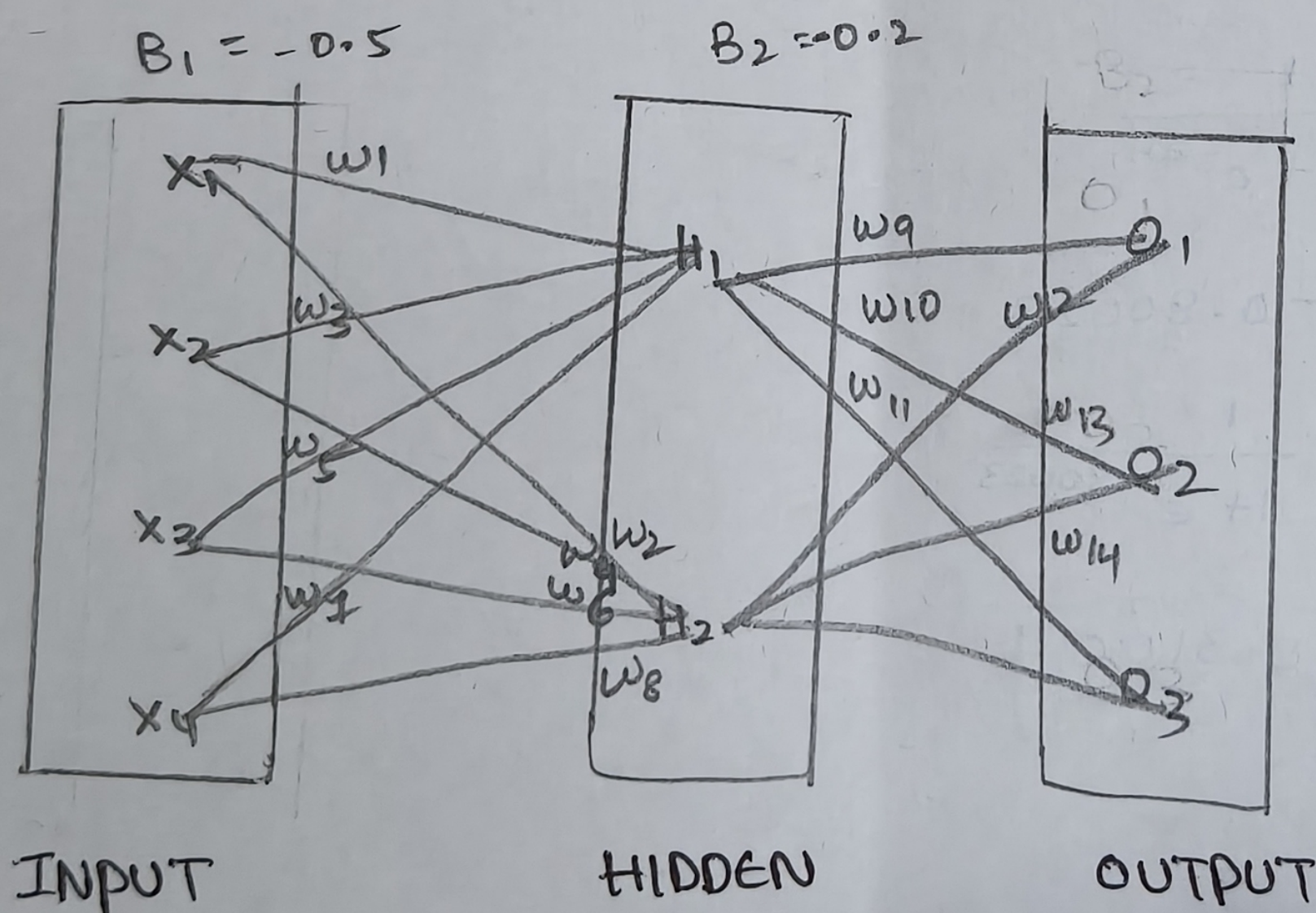
$$X_4 = \text{kernal length}$$

OUTPUT LAYER : 3 neurons

$$O_1 = \text{Kama}$$

$$O_2 = \text{Rosa}$$

$$O_3 = \text{canadian}$$



ONE Row :

$$\text{Input} = [15.26, 14.84, 0.871, 5.763]$$

Weights for H_1 :

$$w_1 = 0.05, w_3 = -0.08, w_5 = 0.01, w_7 = 0.02$$

Weights for H_2 :

$$w_2 = -0.03, w_4 = 0.06, w_6 = 0.008, w_8 = 0.01$$

Bias :

$$\text{Bias } B_1 : -0.5$$

$$\text{Bias } B_2 : -0.2$$

Calculating H_1

$$Z_{H_1} = x_1 w_1 + x_2 w_3 + x_3 w_5 + x_4 w_7 + B_1$$

$$\begin{aligned} Z_{H_1} &= (15.26 \times 0.05) + (14.84 \times -0.08) + (0.871 \times 0.01) + (5.763 \times 0.02) \\ &\quad - 0.5 \\ &= 0.763 - 1.1872 + 0.00871 + 0.11526 - 0.5 \end{aligned}$$

$$Z_{H_1} = -0.80023$$

Sigmoid for H_1

$$H_1 = \frac{1}{1 + e^{-Z_{H_1}}}$$

$$Z_{H_1} = -0.80023$$

$$H_1 = \frac{1}{1 + e^{0.80023}}$$

$$H_1 = 0.3100$$

calculating H_2

$$Z_{H_2} = x_1 w_2 + x_2 w_4 + x_3 w_6 + x_4 w_8 + B_2$$

$$= (15.26 \times -0.03) + (14.84 \times 0.06) + (0.871 \times 0.008) + (5.763 \times 0.01) - 0.2$$

$$Z_{H_2} = 0.2972$$

Sigmoid for H_2

$$H_2 = \frac{1}{1 + e^{-Z_{H_2}}}$$

$$H_2 = \frac{1}{1 + e^{-0.2972}}$$

$$H_2 = 0.5738$$

$$\therefore H = [0.3100, 0.5738]$$

output layer :

$$\text{Kama} : w_9 = 1, w_{10} = -0.2, B_3 = 0$$

$$\text{Rosa} : w_{11} = -0.5, w_{12} = 1, B_4 = -0.1$$

$$\text{canadian} : w_{13} = 0.2, w_{14} = -0.1, B_5 = -0.05$$

$$\text{for Kama } z = H_1 w_9 + H_2 w_{10} + B_3$$

$$Z_{\text{kama}} = (0.3100 \times 1) + (0.5738 \times -0.2) + 0$$
$$= 0.3100 - 0.11476$$

$$Z_{\text{kama}} = 0.1952$$

$$\text{for Rosa } z = H_1 w_{11} + H_2 w_{12} + B_4$$

$$Z_{\text{Rosa}} = (0.3100 \times -0.5) + (0.5738 \times 1) - 0.1$$
$$= -0.155 + 0.5738 - 0.1$$

$$Z_{\text{Rosa}} = 0.3188$$

for canadian $z = H_1 w_{13} + H_2 w_{14} + B_5$

$$\begin{aligned} z_{\text{canadian}} &= (0.3100 \times 0.2) + (0.5738 \times -0.1) - 0.05 \\ &= 0.0620 - 0.05738 - 0.05 \end{aligned}$$

$$z_{\text{canadian}} = -0.0454$$

Final output :

$$\begin{aligned} \text{output} &= [z_{\text{kama}}, z_{\text{rosa}}, z_{\text{canadian}}] \\ &= [0.1952, 0.3188, -0.0454] \end{aligned}$$

Highest score is $= 0.3188$

Prediction : predicted class = Rosa.

Actual : Actual class = kama.

FINAL ANSWER :

ROSA